

VILNA AGCM, AB, Canada

WMO: 717610

Lat: 54.130N

Lon: 111.921W

Elev: 645

StdP: 93.81

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.1	-29.5	-36.7	0.1	-33.0	-32.8	0.2	-29.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	27.4	17.1	25.7	16.6	24.1	15.9	19.1	24.9	17.9	23.7	16.9	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.9	22.6	15.6	12.0	20.9	14.6	11.2	19.6	57.1	24.8	53.1	23.6	49.8	22.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-38.1	30.6	3.1	1.1	-40.3	31.4	-42.1	32.0	-43.9	32.6	-46.1	33.4
			WB	-38.2	21.5	3.1	1.5	-40.4	22.6	-42.2	23.4	-44.0	24.3	-46.2	25.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-12.2	-12.5	-5.7	2.8	10.4	14.4	16.5	15.3	10.5	3.4	-5.8	-13.4
	DBStd	12.49	8.72	7.92	7.69	5.14	4.55	3.03	2.55	3.06	4.40	4.67	6.83	8.23
	HDD10.0	3539	688	631	488	222	52	5	0	2	46	211	473	724
	HDD18.3	5969	946	864	746	467	248	122	69	101	236	464	723	983
	CDD10.0	635	0	0	0	4	62	136	201	166	61	5	0	0
	CDD18.3	24	0	0	0	0	1	3	12	7	1	0	0	0
	CDH23.3	515	0	0	0	1	60	81	172	134	67	1	0	0
	CDH26.7	79	0	0	0	0	9	9	27	19	14	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	437	19	14	18	31	43	76	83	62	36	22	20	19
	PrecMax	526	50	32	35	86	81	162	183	138	92	56	56	36
	PrecMin	278	0	1	5	2	4	22	32	22	3	5	2	1
	PrecStd	67	11	7	9	17	21	33	28	26	22	11	14	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	5.7	11.4	22.1	28.0	28.0	29.5	29.0	29.0	21.3	11.6	5.2
		MCWB	3.1	2.3	5.7	10.2	13.9	16.5	19.5	18.4	15.7	12.3	6.0	1.5
	2%	DB	4.1	3.1	7.9	17.5	25.0	25.6	27.0	26.5	25.1	16.1	7.2	2.3
		MCWB	1.4	0.7	3.4	7.9	12.6	15.6	18.5	17.8	14.8	9.4	3.9	-0.2
	5%	DB	2.3	1.0	5.6	14.3	22.3	23.7	25.2	24.7	22.2	13.4	4.6	0.2
		MCWB	0.3	-0.7	2.2	6.5	11.5	14.7	17.8	17.3	13.8	7.6	2.2	-1.6
	10%	DB	0.1	-0.7	3.6	11.2	20.0	21.8	23.5	23.0	19.2	10.8	2.5	-2.3
		MCWB	-1.4	-2.1	1.0	4.9	10.4	14.1	16.7	16.3	12.4	6.2	0.7	-3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.8	6.3	10.7	15.0	18.6	21.2	20.4	17.2	12.5	6.6	1.9
		MCDB	6.1	5.1	10.7	21.7	24.6	25.3	26.7	26.1	26.2	19.6	10.3	4.6
	2%	WB	1.7	0.9	3.8	8.2	13.7	16.9	19.5	18.9	15.5	10.0	4.2	0.0
		MCDB	3.8	2.7	7.3	16.6	22.1	22.8	25.4	24.4	23.4	15.0	6.8	2.1
	5%	WB	0.3	-0.4	2.4	6.8	12.5	15.8	18.3	17.7	14.2	8.3	2.6	-1.6
		MCDB	2.2	1.0	5.2	13.5	20.9	21.6	24.0	23.2	21.6	12.4	4.5	0.0
	10%	WB	-1.4	-2.2	1.1	5.4	11.2	14.8	17.2	16.6	12.9	6.8	0.7	-3.4
		MCDB	0.1	-0.6	3.5	10.5	18.3	20.0	22.4	22.1	18.5	10.5	2.3	-2.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.6	9.9	10.8	13.6	11.8	12.0	12.6	12.9	10.2	8.4	8.6
		MCDBR	10.0	11.4	11.4	16.2	17.5	15.7	15.2	15.8	18.7	14.9	10.7	10.5
	5% WB	MCWBR	7.9	9.4	7.7	8.7	7.8	7.3	8.0	8.2	9.4	8.8	7.4	8.6
		MCDWR	9.9	10.9	10.9	15.0	15.9	13.1	14.2	14.4	17.7	14.0	10.1	10.3
(40) Clear-Sky Solar Irradiance	taub		0.255	0.264	0.297	0.327	0.343	0.356	0.357	0.365	0.308	0.278	0.250	0.235
		taud	2.296	2.315	2.331	2.380	2.389	2.370	2.407	2.341	2.508	2.562	2.450	2.296
	Ebn at Noon		721	845	887	903	903	891	883	851	872	833	743	675
		Edh at Noon	56	77	96	105	110	114	108	109	80	61	48	45
(44) All-Sky Solar Radiation	RadAvg		0.89	1.93	3.45	4.66	5.71	5.90	6.01	4.91	3.43	1.93	0.91	0.64
	RadStd		0.10	0.15	0.26	0.40	0.46	0.41	0.32	0.37	0.36	0.28	0.11	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page